Kevin Rivera

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EDUCATION

Duke University *August 2026 – December 2027 (Expected)*
Master of Engineering in Robotics & Autonomy
Relevant Coursework: Linear System Theory and Machine Learning, Continuum Robotics, Management in High Tech Industries

University of Central Florida *May 2022 – May 2026*
Bachelor of Science, Mechanical Engineering
GPA: 3.51/4.00
Relevant Coursework: Intro to Robotics, Vibrations & Controls, Heat Transfer 1 & 2, Thermodynamics, CAE Design

EXPERIENCE

Agricultural Robotics Lab – UCF *May 2025 – August 2026*
Volunteer Undergraduate Research Assistant
- Developed a novel, low-cost computer vision depth-estimation system for a strawberry-harvesting robot using two RGB webcams; achieved ~1% error vs. measured distance over a 5-25 cm range without RGB-D cameras, LiDAR, or neural networks
- Designed the real-time MATLAB perception pipeline (RGB decomposition, thresholding, morphological operations, kernel smoothing) that extracts each fruit's perimeter and bottom-edge contour for depth estimation
- Designed modular SolidWorks fixtures and end-effector mounts for section-only reprints, saving ~6 hours per reprint
- Co-authored a conference paper in preparation as primary author; presented results weekly to the lab director

Hubbard Construction (Vinci) *May 2024 – January 2025*
Estimating Intern
- Performed quantity takeoffs from engineering plans for 30+ asphalt estimates ($200K to multimillion) in Excel and Bluebeam Revu; takeoffs supported winning bids including Interstate 4 and the Epic Universe bus depot
- Reconciled takeoffs against engineer-predicted quantities, tracing discrepancies to mislabeled plans or calculation errors

PROJECTS

Observer-Based Fault Detection and Isolation for Industrial Systems | Course project, solo *October 2026 – Present*
- Initiated a Python (python-control, NumPy) state-observer bank to detect and isolate sensor drift and actuator degradation for predictive maintenance

Continuum Robotics – Collapsible SpiRobs Arm | Course project, 4-person team *August 2026 – Present*
- Independently iterated the arm through 5 SolidWorks design-for-manufacturing revisions, fixing cracking, delamination, and undersized holes; optimized geometry to collapse ~40% shorter (CAD estimate)

Wizardry Chess | 3-person team *January 2026 – May 2026*
- Assembled and wired the XY gantry, stepper motors, and electromagnet for an automated chess board playing live Lichess games via API (Raspberry Pi, ROS2); tested at ~2-4 s per move; traced misplaced moves to speech-to-text errors

UCF Senior Design – Bring ROAR to Life | Animatronic Stegosaurus, 5-person team *August 2025 – May 2026*
- Owned electrical design, C++ firmware, and system integration: designed wiring diagrams; soldered and integrated an ESP32 driving 11 servos, an IMU, temperature sensing, and audio
- Diagnosed false tip-over alarms caused by 12 V servo power noise in IMU readings; implemented a 5-sample filter with spike rejection and asymmetric weighting that reduced false fail-safe triggers
- Designed safety systems: tip-over and overheat lockouts with staff alerts, fuses, emergency stops, and fully enclosed electronics
- Built an ESP32-hosted web interface bridging a PS4 controller via smartphone, with a tested ~100-150 ft range
- Delivered the prototype ~44% under budget ($450 of $800) through purchasing decisions; co-authored design reports
- Served as point of contact for the Natural History Museum, London and Lawrence Hall of Science; added hot-swap batteries per their feedback; validated the prototype in 3 demos

LEADERSHIP & ACTIVITIES

Phi Eta Mu – Committee of Medios Electrónicos | Committee Member *September 2026 – Present*
- Contacted 8 employers for the March 2027 career fair (prior fairs: 600-800 attendees, 40+ employers)

Phi Eta Mu – Delta Chapter Non-Profit | Guardian & Former Canciller, Secretary *August 2024 – Present*
- Led a 40-member chapter as Canciller; organized cross-chapter events (80+ attendees), raising attendance ~20%
- Founded the Phi Eta Mu Networking LinkedIn group (110+ members in 5 months); hosted ~7 Zoom career panels
- Awarded the Orden José Menéndez (2026), given to at most 2 members per year since 1981

Duke AERO (Rocketry) | Ground Systems Member • **Duke Advanced Degree Consulting Club** | Consultant (Fall 2026 – Present)

SKILLS

Software: MATLAB, Python (NumPy, python-control), C++, ROS2, SolidWorks, Fusion 360, AutoCAD, ANSYS and Gazebo (coursework), Linux, Git, Microsoft Office, Bluebeam Revu
Technical: Robotics, Computer Vision, Image Processing, Controls, Embedded Systems, System Integration, Prototyping, Design for Manufacturing, Root Cause Analysis, Technical Documentation, Soldering
Hardware: ESP32, Arduino, Raspberry Pi (Pico), Servos, Stepper Motors, IMU | **Languages:** Spanish and English